Automotive Product Profitability Analytics Dashboard

Car Sales Analysis Report

1. Introduction:

The purpose of this project is to analyze a car dataset to:

- Feature demand (popular car attributes).
- Pricing and profitability trends.
- Market performance patterns over time.

The report follows ETL (Extract → Transform → Load) steps, data modeling, DAX calculations, and visualization .

2. Data Cleaning and Preprocessing:

1. **Transform Data**
    - Removed duplicates.
    - Cleaned column names and formatted data types.
2. **Handling Missing Values**
    - Engine HP → replaced null values with median (150–250 range).
    - Engine Cylinders → replaced 0/null with 2–4.
    - Market Category → replaced N/A with "Unknown".

3. **Handle Outliers:**

Handling outliers is the process of identifying data points that significantly deviate from the general pattern of a dataset and then applying a method to address their impact on statistical analysis and machine learning models. This often involves methods like removing the outlier, transforming the data, or capping the extreme values to prevent them from disproportionately influencing results and to improve the reliability of the analysis.

- Outlier Engine HP
- Outlier Engine Cylinder
- Outlier Highway MPG
- Outlier City MPG
- Outlier MSRP
- Outlier Popularity

4. **Normalized Market Category**
    - Split multi-valued categories into rows.
    - Created Dim Market Category and bridge table.

3. Data Transformation and Modeling:

- **Fact Table** → CarFacts (main dataset with prices, MPG, popularity, etc.).
- **Dimension Tables** →
    - DimMake (car manufacturers).
    - DimEngine (HP, cylinders).
    - DimMarketCategory.
    - DimDate (Year, Quarter, Month).

Relationships Created

- CarFacts[CarID] → DimCar[CarID].
- CarFacts[CarID] → Bridge_MarketCategory[CarID].
- Bridge_MarketCategory[MarketCategory] → DimMarketCategory[MarketCategory].
- CarFacts[Year] → DimDate[Year].

4, DAX Measures:

Definition

A DAX measure is a calculation created using DAX (Data Analysis Expressions) in that summarizes or analyzes data, such as totals.

- Total Revenue
- Dim Date
- Total Unit proxy
- Total Profit
- Profit Margin

- Profit Margin Score
- Feature Popularity Index
- Total Sales
- Total Quentity
- Price Elasticity Estimate
- Dim Table
- Rolling 3M Profit
- Rolling 3M Sales

5. Visualizations (Dashboard Pages)

Page 1: Product Feature Demand

- Bar chart compares **types of transmissions** across cars.
- Heatmap shows **average cars by market category (hatchback, crossover, luxury, etc.)**.
- Donut chart shows **distribution of driven wheels** (AWD, FWD, RWD, 4WD).
- Car image with filter panel lets you select **make/year (Acura, BMW, Audi, etc.)**.

Page 2: Pricing Profit Analysis

- Bar chart shows **market category distribution by MSRP**.
- Line chart shows **median car year trend over time**.
- Right panel lists **top car models with quantity**.
- Donut chart shows **maximum price (MSRP) by highway mileage (MPG)**.

Page 3: Market Trend Patterns

- Shows **popularity of cars** by *market category* and *price (MSRP)*.
- Highlights **transmission type usage** (automatic, manual, etc.).
- Map shows **transmission trends by fuel type** in regions.
- Displays **first make, year, model, and vehicle style** for cars (example: Acura 1990).

6. Advanced Analysis

Scenario Slicer

- **Added slicer for:**
    - Electric/Hybrid Vehicles.
    - Traditional Fuel Vehicles.
    - Alternative Fuel Vehicles.

- **Correlation & Pattern Insights**

- Heatmap shows strong relation between Engine HP ↔ MSRP.
- Certain features (Luxury, Crossover) increase both price and popularity.
- Electric/Hybrid show higher MSRP but lower volume compared to traditional cars.

7. Conclusion

- **Feature Demand** → Automatic transmission, SUVs, and Crossovers are most popular.
- **Pricing & Profitability** → Higher HP and luxury segments drive more profit.
- **Market Trends** → Traditional fuel dominates in volume, but electric/alternative vehicles are growing.
- **Next Steps** → Focus on electric vehicle analysis and refine cost assumptions for real margin.